

Memory-Greedy Fictitious Play in Cyclic Three-Action Games

Abstract

We resolve the memory conjecture of Markakis and Santorinaios for exact zero-floor greedy fictitious play in the standard rock–paper–scissors game. From a common pure initial strategy, choosing a tied best response different from the previously selected response gives the unique steps $2/3, 1/2, 1/3, \dots$ and duality gap $2/(3t)$ in skew-symmetric normalization. The complete trajectory is explicit, and it remains unchanged through any horizon $T \geq 2$ when a stepsize floor at most $1/T$ is imposed. For positively weighted cyclic three-action games, we prove that synchronized play from any common non-equilibrium mixed strategy enters a cyclic response pattern after at most one initial zero step. An exact recurrence for the reciprocal payoff gives the asymptotic gap constant $6abc/(a+b+c)^2$, where $a, b, c > 0$ are the cyclic weights. Finally, a four-action example shows that every pure best-response pair can have zero as its unique greedy stepsize, even though a mixed best-response pair leads directly to equilibrium.

1 Introduction

Greedy fictitious play chooses a pair of pure best responses and then minimizes the duality gap along the segment toward that pair. Markakis and Santorinaios [1] study an almost-greedy variant that restricts the stepsize to $[\delta, 1]$ for a positive parameter δ . In Section 5 of their paper, they conjecture that a memory rule resolves the tie-breaking obstruction in the standard three-action rock–paper–scissors game: from a common pure initialization, selecting a different tied best response from the previous iteration should give steps $2/3, 1/2, 1/3, \dots$ and convergence at rate $1/T$.

We prove the exact stepsize assertion for line search over $[0, 1]$ and identify the entire trajectory and its duality gap. An infinite trajectory with steps $1/t$ requires that no fixed positive stepsize floor be imposed. We also show that any floor $0 \leq \delta \leq 1/T$ preserves the trajectory through a prescribed horizon $T \geq 2$.

The same calculation extends to cyclic three-action skew-symmetric games with arbitrary positive weights. From a common non-equilibrium mixed strategy and synchronized response selection, at most the first update can have zero stepsize. Thereafter the responses cycle, and the reciprocal of the maximum payoff has an explicit periodic increment. We derive both a finite-time formula and its asymptotic gap constant. A four-action example then shows why changing pure tie-breaking alone cannot guarantee progress in all games.

Our results concern adaptive greedy steps. For ordinary fictitious play, the general weak Karlin conjecture was disproved by Wang [2]. Related algorithms using mixed directions and the duality gap include the method of Fasoulakis et al. [3].

2 The update rule

For a payoff matrix $A \in \mathbb{R}^{m \times n}$, the row player maximizes $x^\top A y$ and the column player minimizes it. Their mixed strategies belong to the probability simplices Δ_m and Δ_n , where

$$\Delta_d = \left\{ x \in \mathbb{R}^d : x_i \geq 0 \text{ for all } i, \sum_{i=0}^{d-1} x_i = 1 \right\}.$$

We index actions from zero, write e_i for the corresponding unit vector, and use $\mathbf{1}$ for the all-ones vector of the appropriate dimension. The duality gap is

$$\psi_A(x, y) = \max_{0 \leq i < m} (A y)_i - \min_{0 \leq j < n} (x^\top A)_j. \quad (2.1)$$

It is the sum of the two players' unilateral improvement amounts, so it is nonnegative and vanishes exactly at a Nash equilibrium.

Given a current profile (x, y) , select pure best responses

$$\begin{aligned} i &\in \arg \max_{0 \leq r < m} (A y)_r, \\ j &\in \arg \min_{0 \leq s < n} (x^\top A)_s. \end{aligned}$$

The greedy stepsize is any minimizer

$$\eta \in \arg \min_{0 \leq z \leq 1} \psi_A((1-z)x + z e_i, (1-z)y + z e_j). \quad (2.2)$$

Both players use the same stepsize. A stepsize floor $\delta \in [0, 1]$ replaces the search interval by $[\delta, 1]$.

Common states in skew-symmetric games. Suppose $A^\top = -A$ and $x = y$. For each action j ,

$$(x^\top A)_j = -(A x)_j.$$

Thus both players have the same best-response set

$$\text{BR}(x) = \arg \max_i (A x)_i.$$

Define the payoff height $h(x) = \max_i (A x)_i$. Skew-symmetry also gives

$$\psi_A(x, x) = \max_i (A x)_i - \min_j (-(A x)_j) = 2h(x). \quad (2.3)$$

Selecting the same response for both players preserves a common state. Consequently, minimizing the duality gap along a common direction is equivalent to minimizing the height.

The memory rule. Start with $x_0 = y_0$ and stop whenever the gap is zero. At the first update, choose a common response $i_1 \in \text{BR}(x_0)$. At update $t \geq 2$, choose a common response according to

$$i_t \in \begin{cases} \text{BR}(x_{t-1}) \setminus \{i_{t-1}\}, & \text{if this set is nonempty,} \\ \text{BR}(x_{t-1}), & \text{otherwise.} \end{cases} \quad (2.4)$$

Any remaining choice is resolved identically for both players. Use (2.2) with $i = j = i_t$ to obtain η_t , and set

$$x_t = y_t = (1 - \eta_t)x_{t-1} + \eta_t e_{i_t}. \quad (2.5)$$

Record i_t even when $\eta_t = 0$. The index t counts every update, including an update with zero stepsize. Unless a floor is explicitly specified, the line search is over $[0, 1]$.

3 The exact rock–paper–scissors trajectory

Action indices in three-action games are taken modulo three. Consider the skew-symmetric payoff matrix

$$A = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}. \quad (3.1)$$

Thus action $k + 1$ beats action k .

Theorem 1 (Exact trajectory). *For (3.1), run the memory rule from $x_0 = y_0 = e_0$. Each line search has a unique minimizer. The selected response is $i_t = t \pmod{3}$, and the stepsizes are*

$$\eta_t = \begin{cases} \frac{2}{3}, & t = 1, \\ \frac{1}{t}, & t \geq 2. \end{cases} \quad (3.2)$$

For every $t \geq 1$, the common state and its gap satisfy

$$x_t = y_t = \frac{1}{3}\mathbf{1} + \frac{e_{i_t} - e_{i_t+1}}{3t}, \quad (3.3)$$

$$\psi_A(x_t, y_t) = \frac{2}{3t}. \quad (3.4)$$

Cyclic relabeling gives the corresponding statement from any common pure initial strategy.

Proof. The first update. At $x_0 = e_0$, the payoff vector is $(0, 1, -1)^\top$, so action 1 is the unique best response. Along the segment to e_1 ,

$$A((1 - \eta)e_0 + \eta e_1) = \begin{pmatrix} -\eta \\ 1 - \eta \\ 2\eta - 1 \end{pmatrix}.$$

The first component is always smaller than the second. The height is therefore $\max\{1 - \eta, 2\eta - 1\}$. One affine function strictly decreases and the other strictly increases; they intersect inside $[0, 1]$ at the solution of

$$1 - \eta = 2\eta - 1.$$

The unique minimizing step is $\eta_1 = 2/3$. Direct substitution gives

$$x_1 = \begin{pmatrix} 1/3 \\ 2/3 \\ 0 \end{pmatrix} = \frac{1}{3}\mathbf{1} + \frac{e_1 - e_2}{3},$$

$$Ax_1 = \begin{pmatrix} -2/3 \\ 1/3 \\ 1/3 \end{pmatrix}.$$

This proves the state and gap formulas at $t = 1$; the new best-response set is exactly $\{1, 2\}$.

The induction step. Suppose (3.3) holds at update $t \geq 1$, with $i_t = t \pmod{3}$. Write $j = i_t$, $k = j + 1$, $l = j + 2$, and $h = 1/(3t)$. The identities $A\mathbf{1} = 0$ and $A(e_j - e_k) = e_j + e_k - 2e_l$ imply

$$Ax_t = h(e_j + e_k - 2e_l).$$

Hence $\text{BR}(x_t) = \{j, k\}$, and the memory rule must select $i_{t+1} = k$. For a proposed step η , define $u(\eta) = (1 - \eta)x_t + \eta e_k$. Its three payoff components are

$$\begin{aligned} (Au(\eta))_j &= (1 - \eta)h - \eta, \\ (Au(\eta))_k &= (1 - \eta)h, \\ (Au(\eta))_l &= -2(1 - \eta)h + \eta. \end{aligned} \quad (3.5)$$

The j component never exceeds the k component and is strictly smaller for $\eta > 0$. The k component has slope $-h < 0$, whereas the l component has slope $1 + 2h > 0$. Their difference is

$$(Au(\eta))_l - (Au(\eta))_k = (1 + 3h)\eta - 3h.$$

Thus the height strictly decreases up to their crossing and strictly increases after it. The unique minimizer is

$$\eta_{t+1} = \frac{3h}{1 + 3h} = \frac{1}{t + 1}. \quad (3.6)$$

At this step, precisely the k and l components attain the height

$$h(x_{t+1}) = (1 - \eta_{t+1})h = \frac{1}{3(t + 1)}.$$

For completeness, the state update can be simplified directly:

$$\begin{aligned} x_{t+1} &= \frac{t}{t + 1} \left(\frac{1}{3} \mathbf{1} + \frac{e_j - e_k}{3t} \right) + \frac{1}{t + 1} e_k \\ &= \frac{t}{3(t + 1)} \mathbf{1} + \frac{e_j + 2e_k}{3(t + 1)} \\ &= \frac{1}{3} \mathbf{1} + \frac{e_j + 2e_k - \mathbf{1}}{3(t + 1)} \\ &= \frac{1}{3} \mathbf{1} + \frac{e_k - e_l}{3(t + 1)}. \end{aligned}$$

The last equality uses $\mathbf{1} = e_j + e_k + e_l$. This is the required state formula at $t + 1$. Equation (2.3) gives the gap and completes the induction. $\square$

Corollary 2 (Stepsize floors and normalization). *Fix an integer $T \geq 2$. If every update instead minimizes over $[\delta, 1]$ with $0 \leq \delta \leq 1/T$, Theorem 1 holds through update T . For the normalized matrix*

$$\tilde{A} = \frac{1}{2}(A + \mathbf{1}\mathbf{1}^\top),$$

the responses, stepsizes, and states are unchanged, and the gap is

$$\psi_{\tilde{A}}(x_t, y_t) = \frac{1}{3t}. \quad (3.7)$$

Proof. For $T \geq 2$, the first step $2/3$ and all steps $1/t$ with $2 \leq t \leq T$ are at least $1/T$. A unique minimizer on $[0, 1]$ remains the unique minimizer on any subinterval containing it. Inducting over the updates therefore preserves both the state and the recorded response through time T .

For normalization, every probability vector satisfies $\mathbf{1}^\top x = \mathbf{1}^\top y = 1$. Consequently,

$$\begin{aligned} (\tilde{A}y)_i &= \frac{1}{2}(Ay)_i + \frac{1}{2}, \\ (x^\top \tilde{A})_j &= \frac{1}{2}(x^\top A)_j + \frac{1}{2}. \end{aligned}$$

These transformations preserve both best-response sets. Subtracting the column minimum from the row maximum gives $\psi_{\tilde{A}}(x, y) = \psi_A(x, y)/2$ for every profile. The line-search minimizers are therefore unchanged, and (3.7) follows from (3.4). $\square$

Remark 3 (The role of the floor). A fixed $\delta > 0$ excludes the step $1/t$ as soon as $t > 1/\delta$. Thus the corollary concerns a prescribed finite horizon, whereas the infinite-time trajectory in Theorem 1 uses a zero floor.

Why memory matters. At x_1 , repeating response 1 would give

$$A((1 - \eta)x_1 + \eta e_1) = \begin{pmatrix} -2/3 - \eta/3 \\ 1/3 - \eta/3 \\ 1/3 + 2\eta/3 \end{pmatrix}.$$

The third component is maximal for every $\eta \in [0, 1]$ and strictly increases. The unique minimizing step is therefore zero. Repeated lexicographic selection of action 1 stalls at gap $2/3$, while the memory rule selects action 2 and takes step $1/2$.

4 Positively weighted cyclic games

Let $a, b, c > 0$ and consider

$$A = \begin{pmatrix} 0 & -a & b \\ a & 0 & -c \\ -b & c & 0 \end{pmatrix}. \quad (4.1)$$

Set $S = a + b + c$ and define the probability vector

$$\pi = \frac{1}{S} \begin{pmatrix} c \\ b \\ a \end{pmatrix}.$$

Direct multiplication gives $A\pi = 0$ and $\pi^\top A = 0$. Moreover, $Ax = 0$ implies $ax_0 = cx_2$ and $ax_1 = bx_2$; normalization then gives $x = \pi$. For any common state x , the payoffs $s = Ax$ satisfy

$$\sum_{r=0}^2 \pi_r s_r = 0. \quad (4.2)$$

Because every π_r is positive, a nonzero payoff vector has both a positive and a negative component. Thus $h(x) > 0$ whenever $x \neq \pi$, and π is the unique common state with zero gap. In particular, it is the unique symmetric equilibrium.

The next lemma isolates the line-search calculation used throughout this section.

Lemma 4 (A single weighted update). *Let $x \neq \pi$ be a common state, let $s = Ax$, and put $h = \max_r s_r$. Select any $k \in \text{BR}(x)$, and write $l = k + 1$, $j = k - 1$, and $\gamma_k = A_{lk} > 0$. The unique minimizing step toward e_k is*

$$\eta^* = \frac{h - s_l}{h - s_l + \gamma_k}. \quad (4.3)$$

It is zero exactly when $s_l = h$. If $s_l < h$, then $0 < \eta^ < 1$ and the new common state $x^+ = (1 - \eta^*)x + \eta^* e_k$ satisfies*

$$\text{BR}(x^+) = \{k, l\}, \quad (4.4)$$

$$h(x^+) = (1 - \eta^*)h > 0. \quad (4.5)$$

Proof. For $u(\eta) = (1 - \eta)x + \eta e_k$, define $f_r(\eta) = (Au(\eta))_r$. Since $s_k = h$ and $A_{kk} = 0$,

$$\begin{aligned} f_k(\eta) &= (1 - \eta)h, \\ f_j(\eta) &= (1 - \eta)s_j + \eta A_{jk}, \\ f_l(\eta) &= (1 - \eta)s_l + \eta \gamma_k. \end{aligned} \quad (4.6)$$

The sign $A_{jk} < 0$ gives the following inequality, which is strict whenever $\eta > 0$:

$$f_k(\eta) - f_j(\eta) = (1 - \eta)(h - s_j) - \eta A_{jk} \geq 0.$$

Hence the height along the segment equals $\max\{f_k(\eta), f_l(\eta)\}$.

The function f_k strictly decreases because $h > 0$. To see that f_l strictly increases, note that γ_k is the unique largest entry in row l of A . The value s_l is a convex combination of that row's entries, so $s_l \leq \gamma_k$, with equality only when $x = e_k$. But at e_k , action l earns $\gamma_k > 0$ and action k earns zero, contradicting $k \in \text{BR}(x)$. Therefore $s_l < \gamma_k$, which is precisely the positivity of the slope of f_l .

If $s_l < h$, the two functions satisfy $f_k(0) > f_l(0)$ and $f_k(1) = 0 < f_l(1) = \gamma_k$. Their crossing is the unique minimizer. Solving the crossing equation yields

$$\begin{aligned}(1 - \eta)(h - s_l) &= \eta\gamma_k, \\ h - s_l &= \eta(h - s_l + \gamma_k),\end{aligned}$$

which gives (4.3). Its numerator is positive and its denominator is larger by $\gamma_k > 0$, so the minimizer lies in $(0, 1)$. At the crossing, the j component is strictly smaller, proving (4.4). The common maximum is $(1 - \eta^*)h > 0$, proving (4.5).

If $s_l = h$, the functions agree at $\eta = 0$. For every $\eta > 0$, $f_l(\eta) > f_l(0) = h$, so zero is the unique minimizer. This also agrees with (4.3). $\square$

Theorem 5 (Weighted cyclic convergence). *For (4.1), run the synchronized memory rule from any common initial state $x_0 = y_0 \neq \pi$. Each line search has a unique minimizer. The first positive update occurs at a time $\tau \in \{1, 2\}$, and all later steps are positive. For $t \geq \tau$,*

$$\text{BR}(x_t) = \{i_t, i_t + 1\}, \quad (4.7)$$

$$i_{t+1} = i_t + 1 \pmod{3}. \quad (4.8)$$

Write $h_t = h(x_t)$, and for each action k set $\alpha_k = \pi_{k+1}A_{k+1,k}$. For $t \geq \tau$ and $k = i_{t+1}$, the exact recurrences are

$$\eta_{t+1} = \frac{h_t}{h_t + \alpha_k}, \quad (4.9)$$

$$\frac{1}{h_{t+1}} = \frac{1}{h_t} + \frac{1}{\alpha_k}. \quad (4.10)$$

In particular,

$$\frac{1}{h_{t+3}} = \frac{1}{h_t} + \frac{S^2}{abc} \quad (t \geq \tau), \quad (4.11)$$

and

$$\lim_{t \rightarrow \infty} t \psi_A(x_t, x_t) = \frac{6abc}{S^2}. \quad (4.12)$$

Proof. Entry into the cyclic pattern. At a common state $x \neq \pi$, put $s = Ax$ and $h = \max_r s_r > 0$. If all three actions maximized, then $s_0 = s_1 = s_2 = h$, contradicting (4.2). Thus there are at most two best responses.

Apply Lemma 4 to the initial response $k = i_1$. If its successor $l = k + 1$ has score below h , the first step is positive and the new best-response set is $\{k, l\}$. If $s_l = h$, the first step is zero, and the best-response set at the unchanged state is exactly $\{k, l\}$. Because the recorded response is k , memory next selects l . The successor of l is $j = k - 1$, whose score is strictly below h , so the second step is positive. This proves $\tau \in \{1, 2\}$.

After any positive update with response k , the new best-response set is exactly $\{k, k + 1\}$ and its height remains positive. Memory selects $k + 1$ next, and the remaining action has strictly smaller payoff. The lemma therefore applies again with a positive step. Induction proves (4.7)–(4.8) and also shows that no later state is an equilibrium. The uniqueness of every step follows from the lemma, including a possible initial zero step.

The one-step recurrence. Fix $t \geq \tau$. Set $k = i_{t+1}$, $j = k - 1$, $l = k + 1$, and $\gamma_k = A_{lk} > 0$. Immediately before the update, actions j and k both have payoff h_t . Substitution in (4.2) gives

$$\begin{aligned} 0 &= (\pi_j + \pi_k)h_t + \pi_l(Ax_t)_l \\ &= (1 - \pi_l)h_t + \pi_l(Ax_t)_l. \end{aligned}$$

Consequently,

$$\begin{aligned} (Ax_t)_l &= -\frac{1 - \pi_l}{\pi_l}h_t, \\ h_t - (Ax_t)_l &= \frac{h_t}{\pi_l}. \end{aligned}$$

Using this difference in (4.3) yields

$$\eta_{t+1} = \frac{h_t/\pi_l}{h_t/\pi_l + \gamma_k} = \frac{h_t}{h_t + \pi_l\gamma_k}.$$

Since $\alpha_k = \pi_l\gamma_k$, this is (4.9). Equation (4.5) now gives

$$\begin{aligned} h_{t+1} &= \frac{\alpha_k h_t}{h_t + \alpha_k}, \\ \frac{1}{h_{t+1}} &= \frac{h_t + \alpha_k}{\alpha_k h_t} = \frac{1}{h_t} + \frac{1}{\alpha_k}, \end{aligned}$$

which proves (4.10).

The cycle increment and the limit. The three positive constants are

$$\begin{aligned} \alpha_0 &= \pi_1 A_{10} = \frac{ab}{S}, \\ \alpha_1 &= \pi_2 A_{21} = \frac{ac}{S}, \\ \alpha_2 &= \pi_0 A_{02} = \frac{bc}{S}. \end{aligned} \tag{4.13}$$

Any three consecutive responses after time τ use each action once. Summing (4.10) over those updates gives

$$\frac{1}{h_{t+3}} - \frac{1}{h_t} = \frac{S}{ab} + \frac{S}{ac} + \frac{S}{bc} = \frac{S(a+b+c)}{abc} = \frac{S^2}{abc}.$$

This proves (4.11).

Put $C = S^2/(abc)$. For $t \geq \tau$, write $t = \tau + 3q + r$ with integers $q \geq 0$ and $r \in \{0, 1, 2\}$. Repeated use of the recurrence gives the exact formula

$$\frac{1}{h_t} = \frac{1}{h_\tau} + qC + \sum_{u=1}^r \frac{1}{\alpha_{i_\tau+u}}. \tag{4.14}$$

The sum is zero when $r = 0$, and its indices are interpreted modulo three. Substituting $q = (t - \tau - r)/3$ rewrites this as

$$\frac{1}{h_t} = \frac{C}{3}t + \beta_r,$$

where the three fixed offsets are

$$\beta_r = \frac{1}{h_\tau} + \sum_{u=1}^r \frac{1}{\alpha_{i_\tau+u}} - \frac{C}{3}(\tau + r).$$

Because r has only three possible values, $\beta_r/t \rightarrow 0$ along the whole sequence. Using (2.3), we obtain

$$t \psi_A(x_t, x_t) = \frac{2t}{h_t^{-1}} = \frac{2}{C/3 + \beta_r/t} \rightarrow \frac{6}{C} = \frac{6abc}{S^2}.$$

□

For $a = b = c = 1$, the limit is $2/3$, agreeing with Theorem 1. In that theorem, $\tau = 1$ and $h_\tau = 1/3$; indeed, (4.10) then adds exactly 3 to the reciprocal height at every update. The weighted theorem also covers common mixed initial states. Synchronizing the initial response when it is tied is part of its hypothesis: independent initial selections need not preserve a common state.

5 A limitation of pure response selection

The preceding proofs depend on the geometry of a cyclic three-action game. With four actions, even changing both selected pure best responses may fail to produce a direction of decrease.

Proposition 6 (Stalling for every pure pair). *For $0 < \rho < 1$, define*

$$A_\rho = \begin{pmatrix} 0 & -1 & 1 & \rho \\ 1 & 0 & -1 & \rho \\ -1 & 1 & 0 & \rho \\ -\rho & -\rho & -\rho & 0 \end{pmatrix}. \quad (5.1)$$

At $x = y = e_3$, both best-response sets are $\{0, 1, 2\}$. For any selected pair $i, j \in \{0, 1, 2\}$, put

$$\begin{aligned} x_i(\eta) &= (1 - \eta)e_3 + \eta e_i, \\ y_j(\eta) &= (1 - \eta)e_3 + \eta e_j. \end{aligned}$$

For every $\eta \in [0, 1]$,

$$\psi_{A_\rho}(x_i(\eta), y_j(\eta)) = 2\rho + 2(1 - \rho)\eta. \quad (5.2)$$

Thus zero is the unique greedy stepsize for every pure best-response pair, including pairs with $i \neq j$.

In contrast, the common mixed best response

$$p = \begin{pmatrix} 1/3 \\ 1/3 \\ 1/3 \\ 0 \end{pmatrix}$$

gives, for $u(\eta) = (1 - \eta)e_3 + \eta p$,

$$\psi_{A_\rho}(u(\eta), u(\eta)) = 2\rho(1 - \eta). \quad (5.3)$$

It therefore reaches equilibrium at the unique minimizing step $\eta = 1$.

Proof. Initially,

$$\begin{aligned} A_\rho e_3 &= (\rho, \rho, \rho, 0)^\top, \\ e_3^\top A_\rho &= (-\rho, -\rho, -\rho, 0). \end{aligned}$$

These vectors give the asserted best-response sets. Let B be the upper-left 3×3 block of A_ρ . Every column of B contains $-1, 0, 1$, as does every row. For a pure column direction $j \in \{0, 1, 2\}$,

$$\begin{aligned} (A_\rho y_j(\eta))_r &= (1 - \eta)\rho + \eta B_{rj} & (r \in \{0, 1, 2\}), \\ (A_\rho y_j(\eta))_3 &= -\eta\rho. \end{aligned}$$

The maximum among the first three components is $(1 - \eta)\rho + \eta$. It is positive and exceeds the fourth component for every $\eta \in [0, 1]$. Thus

$$\max_r (A_\rho y_j(\eta))_r = (1 - \eta)\rho + \eta.$$

Likewise, for a pure row direction $i \in \{0, 1, 2\}$,

$$\begin{aligned}(x_i(\eta)^\top A_\rho)_s &= -(1 - \eta)\rho + \eta B_{is} & (s \in \{0, 1, 2\}), \\ (x_i(\eta)^\top A_\rho)_3 &= \eta\rho.\end{aligned}$$

The minimum among the first three components is $-(1 - \eta)\rho - \eta < 0$, so the fourth component cannot attain the minimum. Subtracting the two extremal payoffs gives

$$\psi_{A_\rho}(x_i(\eta), y_j(\eta)) = ((1 - \eta)\rho + \eta) - (-(1 - \eta)\rho - \eta) = 2\rho + 2(1 - \rho)\eta.$$

Since $1 - \rho > 0$, this is strictly increasing, proving the pure-direction claim.

The vector p is supported on initial best responses, so it is itself a mixed best response for either player. The first three components of $A_\rho p$ are zero and the fourth is $-\rho$. Consequently,

$$A_\rho u(\eta) = \begin{pmatrix} (1 - \eta)\rho \\ (1 - \eta)\rho \\ (1 - \eta)\rho \\ -\eta\rho \end{pmatrix}.$$

Its maximum is $(1 - \eta)\rho$. Skew-symmetry and (2.3) give (5.3), which strictly decreases to zero at $\eta = 1$. $\square$

Every rule that uses exact zero-floor line search and changes only the chosen pure best-response pair therefore remains at the initial profile of Proposition 6, with positive gap 2ρ . This example does not address algorithms with a positive stepsize floor or with mixed response directions.

Use of AI tools. OpenAI’s Codex (GPT-6 Astra) was used during proof development, computational checking, and preparation of the manuscript.

References

- [1] Evangelos Markakis and Christodoulos Santorinaios. Do Not Discretize, Optimize: Almost Greedy Fictitious Play. In *Proceedings of the Thirty-Fifth International Joint Conference on Artificial Intelligence (IJCAI 2026)*, pages 3505–3513, 2026. <https://doi.org/10.24963/ijcai.2026/390>. Extended version: arXiv:2606.12149v1, <https://arxiv.org/abs/2606.12149>.
- [2] Yuanhao Wang. *Tie-breaking Agnostic Lower Bound for Fictitious Play*. arXiv:2507.09902, 2025. <https://arxiv.org/abs/2507.09902>.
- [3] Michail Fasoulakis, Evangelos Markakis, Georgios Roussakis, and Christodoulos Santorinaios. A descent-based method on the duality gap for solving zero-sum games. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence (IJCAI 2025)*, pages 3839–3847, 2025. <https://doi.org/10.24963/ijcai.2025/427>. Extended version: arXiv:2501.19138, <https://arxiv.org/abs/2501.19138>.